Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Convergent Manufacturing of Hybrid Metal-Composite Cryogenic Tubes for Liquid Hydrogen Storage and Transportation*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Technology Field 2 is optional*	
	Medium Category	Small Category	Medium Category	Small Category
	New &Renewable Energy	*Hydrogen*		
Concept of Proposed Technology	The transition to a hydrogen-based economy requires a paradigm shift in transportation infrastructure. Current state-of-the-art (SOA) technologies, such as heavy stainless steel or conventional filament-wound composite tubes, struggle with weight and embrittlement in cryogenic environments (-253°C). Metal-composite tubes solve this by fusing a zero-permeation metal liner (e.g., 6061-T6 or Ti-6Al-4V) with a light-weight structural composite overwrap, reducing thermal mass ten-fold and enabling chill-down in under 2 seconds [1]. Traditional filament winding is restricted in shape and in axial fiber angle due to slippage. This limits the axial stiffness needed for long cryogenic pipes or arrays of high aspect hydrogen storage cells that better fit into aerodynamic systems like blended wing bodies. Continuous Fiber 3D Printing (CF3D) enables a fuller range of axial fiber angles, shape control, and radii as tight as 3 cm [2]. We propose to investigate the processing-structure-property relationships of metal-composite tubes with carbon fiber composites having AM tow paths optimized to handle extreme 700 bar pressure and a rapidly curing and low-viscosity (10–300 cP) DCPD resin, which shows a 30% increase in fatigue strength at *–196 °C* compared to room temperature [3]. The composite will be combined with aerogel blankets (e.g., Cryogel® Z) *[4]* which possess the lowest thermal conductivity of any solid due to their nanoporous structure and are capable of desorption cooling, which eliminates hydrogen boil-off. The outer layer will consist of conformal metal layers to enhance robustness, decrease permeability, and enable easy structural attachment. Due to the compliance of the assembly before curing and consolidation via Frontal Polymerization (FP), we anticipate increased shape moldability before creating a chemically anchored interface that prevents delamination during			

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Rationale for Joint R&D Project	extreme thermal cycling. To achieve global decarbonization by 2030, this joint R&D project addresses the critical infrastructure gap in liquid hydrogen storage and transport. While traditional steel and filament-wound systems yield to hydrogen embrittlement and geometric constraints, this collaboration leverages UIUC's pioneering leadership in Frontal Polymerization (FP) and Continuous Fiber 3D Printing (CF3D) to create non-geodesic, high-performance architectures. These innovations eliminate manufacturing voids and delamination, ensuring structural integrity at -253°C and 700 bar pressures. Strategic alignment is anchored in South Korea's designation of hydrogen as a National Strategic Technology. The partnership bridges UIUC's advanced manufacturing with the Korean SME's industrial excellence in aerogel production and precision metal forming (ex. Jios Aerogel, Young). This synergy enables the rapid scaling of indigenous, high internal pressure components such as transfer tubes and conformable tanks for the mobility market. By distributing technical risks and combining U.S. aerospace-grade composite technology with Korea's agile manufacturing base, the project de-risks the transition to a global hydrogen economy and ensures a direct pathway to commercialize zero-boil-off storage solutions.
Keywords	1. **Continuous Fiber 3D Printing (CF3D):** This additive manufacturing process enables continuous "fiber steering" along non-geodesic paths, allowing for the strategic placement of reinforcements according to principal stress vectors, which is impossible with traditional filament winding. 2. **Nano-Silica Aerogel Thermal Insulation:** Integrated nanoporous blankets utilize the Knudsen effect to suppress gas-phase heat transfer, achieving ultra-low thermal conductivity (<0.016 W/mK) to maintain liquid hydrogen stability and minimize boil-off rates. 3. **Cryogenic Hydrogen Storage and Transport (-253°C):** The system is specifically engineered to handle hydrogen in both gaseous (at 700 bar) and liquid states at extreme cryogenic temperatures, meeting the technical requirements for zero-permeation and high-pressure service. 4. **Dicyclopentadiene (DCPD) Frontal Polymerization:** A high-toughness, ultra-low viscosity resin system that utilizes rapid, energy-efficient in-situ curing waves to achieve void-free consolidation and superior cryogenic fracture toughness.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 2	End of the Project	Level 6

R&D Objectives

The overall goal of this research is to develop, demonstrate, and validate a single-stage manufacturing process for hybrid metal-composite tubular structures that overcomes the thermomechanical limitations of SOA filament-wound tanks. This will be achieved by integrating three distinct technologies CF3D, chemical bonding, and Frontal Polymerization into a unified production line.

Explicit Connection to Past Work

This proposal is deeply rooted in the prior successful research of the investigative team, ensuring a high Technology Readiness Level (TRL) at the project's onset.

CF3D Composite Tubes: The team has previously demonstrated the foundational technology for Continuous Fiber 3D Printing of tubular structures. We have successfully fed 12K Toray T1100 carbon fiber tows into a proprietary Fiber Wet-Out Module (FWM), impregnated them with proprietary resin, and printed them on a mandrel. This work validated the ability to snap-cure the resin to a B-stage (degree of cure α=17±5%) immediately upon deposition, creating a pliable preform that can be molded into various shapes with an optimized fiber structure. Seams are minimized through a staggered co-cured ply-drop scheme that has previous been shown to maintain structural integrity. This capability is the prerequisite for achieving the non-geodesic winding angles proposed here.

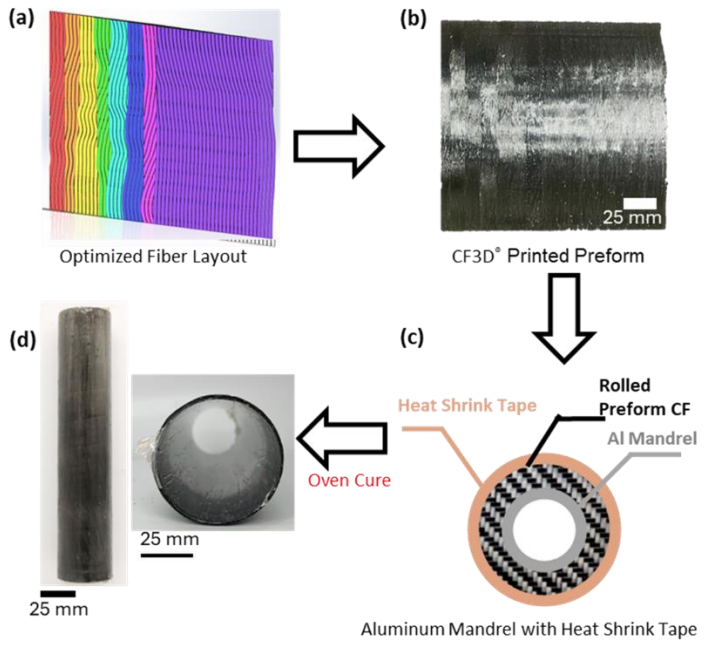

Fig 1. Composite tubes fabricated via our CF3D [2]

Frontal Polymerization and In-Space Manufacturing Integration

The project incorporates transformative research on the Frontal Polymerization (FP) of Dicyclopentadiene (DCPD). This low-energy manufacturing process published in AIAA utilizes a self-propagating thermal front to cure resins, requiring orders of magnitude less energy than conventional oven or autoclave methods and enables rapid manufacturing in austere environments and to meet surge demand. By utilizing high-toughness, low-viscosity DCPD resin systems, the project builds upon the success of DARPA-funded initiatives involving stowable composite trusses. This approach ensures that the resulting cryogenic components benefit from both the structural efficiency of 3D

printing and the extreme energy efficiency of frontal polymerization

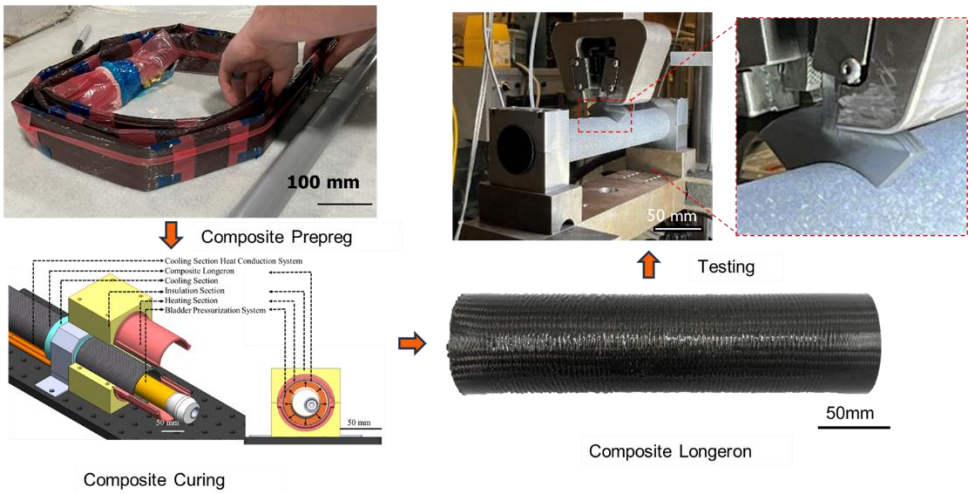

Fig 2. Composite tubes fabricated via DCPD infusion [5]

Objective 1: Formulation of a Covalent Interfacial Bonding Protocol

To eliminate the mechanical interlocking mechanism used in current hybrid tubes (which fails under cryogenic cycling), we will develop a surface functionalization protocol for Aluminum 6061-T6.

Selection: Al 6061 offers optimal specific strength, cryogenic ductility, and low hydrogen embrittlement compared to high-strength steels.

Thickness: The 0.5 - 1.0 mm profile acts as a permeation barrier and minimizes thermal mass to enable the 2-second chill-down.

Target: Achieve a chemical bond strength >20MPa that allows for cohesive failure of the resin rather than adhesive failure at the interface [6].

Approach: Utilizing organofunctional silanes (e.g., A-1387) that bridge the inorganic metal oxide layer and the organic DCPD polymer matrix [7].

Objective 2: Computational Design of non-geodesic via CF3D

Design and analysis of filament-wound composite pressure vessels based on non-geodesic winding

We aim to exploit the steering capabilities of CF3D to fabricate preforms with non-geodesic winding [8].

Target

Objective: Engineer a high-performance composite tube capable of withstanding a burst pressure ≥700 bar (approx. 10,150 psi) using a roll-wrap mandrel approach and metallic end caps.

- Primary Metric: Burst Pressure ≥ 700 bar.
- Design Constraint: Non-geodesic fiber architecture
- Process: CF3D (Continuous Fiber 3D Printing) with UV-curable resin on a mandrel and potential high toughness DCPD resin interlayer, if needed.
- Failure Mode Prevention: Prevent matrix weeping and premature delamination by shifting the load path directly to the carbon fiber filaments.

Approach

Fiber Architecture Selection

- The approach utilizes a non-geodesic ±75° helical winding pattern.
- Rationale: In a tube (with open or non-integrated ends), the hoop stress is the dominant force. A 75° angle is fiber-dominated for axial loads.
- The Innovation: Traditional filament winding would suffer from fiber slip at this angle. Using

R&D Target Specifications

CF3D, the UV-LED system cures the resin at the point of contact, anchoring the fiber to the mandrel instantly and allowing for this steep, high-pressure-resistant angle.

Objective 3: Single-Stage Consolidation via Frontal Polymerization

Demonstrate the feasibility of curing the thick-section hybrid assembly (Metal + Composite + Aerogel insulation) using a single thermal trigger.

Target: Achieve a composite void content of <1% in the final part.
Enabler: The ultra-low viscosity of DCPD (10cP vs 150 cP for epoxy) enables infusion into dense non-geodesic weaves, while the exothermic FP wave ensures uniform cure through the thickness without thermal gradients that cause warping.

Toughness: Fracture toughness (G_{IC}) of DCPD is ~1.7 N/mm, vastly superior to epoxy 0.5 N/mm, making it resilient to cryogenic micro-cracking [3]

Objective 4: Validation of Cryogenic Performance and Passive Cooling

Characterize the thermal and mechanical behavior of the integrated system at -253°C.

Target: Validate the desorption cooling effect of the aerogel layer, aiming for a measurable reduction in tank wall temperature during simulated ascent venting [9].
Material: Cryogel Z (Aspen Aerogels) reinforced with glass fiber batting.
Thickness: 10 mm - 20 mm (2 layers).
Mechanism: Nanoporous silica aerogel d_{pore} = 20 nm suppresses gas convection (Knudsen effect).
Testing: Utilizing UIUC's Cryostat-100 facilities to measure boil-off rates and effective thermal conductivity (K_e) under operational conditions.

We have established aggressive, yet achievable quantitative benchmarks derived from the unique material properties of the DCPD-CF3D system.

Specification	Target Value	Reasoning/Derivation
Interlaminar Fracture Toughness (G_{IC})	≥1.7 N/mm	Material properties of DCPD (ex. Proxima) resin vs. standard Epoxy (0.5 N/mm). This metric ensures resistance to cryo-cracking.
Thermal Mass Reduction	10 x vs. 316L SS	Composite density (1.6 g/cc) is 1/5th of steel (8 g/cc), and wall thickness is reduced due to higher specific strength. Reduced C_p X mass leads to faster thermal equilibrium.
Composite Void Content	< 1.0	Achievable via vacuum-assisted injection of 10 cP DCPD. Standard aerospace composites target < 2%; we target < 1% to minimize H_2 permeability.
Liner-Composite Bond Strength	> 25 MPa	Shear strength of silane-modified Al-Epoxy interfaces typically reaches 20-30 MPa. We target the upper end using optimized surface chemical etching.
Thermal Conductivity (k_{eff})	< 15 mW/m-K	Performance of CryogelZ at ambient pressure. Essential to maintain boil-off < 1.5%/day.

	Operational Pressure	5 - 30 bar	Standard range for LH_2 transfer lines.

R&D Plan

The project is structured into a 3-year timeline, divided into the **Seed Phase (6 months)** for planning and the **Full Phase (2.5 years)** for execution.

Phase 1: Preliminary, Planning & Feasibility (Months 1-6)

Task 1.1: Preliminary Design of Preform Fiber Architecture's Mechanical Behavior

Activity: Perform printing of fiber architecture and understand the mechanical behavior and shape moldability in a mandrel of various diameter.

Deliverable: Optimized winding angle/path file (G-code) for CF3D printing.

Task 1.2: Preliminary Surface Chemistry Formulation

Activity: Screen silane coupling agents (e.g. γ-MPS vs A-187) for compatibility with Al 6061 and DCPD monomer. Conduct lap-shear tests on small coupons.

Deliverable: SOP for metal liner surface preparation.

Task 1.3: Partner Alignment

Activity: Finalize work packages with Korean SME partners (e.g., JIOS Aerogel for insulation, liner manufacturers). Establish IP sharing and export control protocols.

Deliverable: Joint R&D Agreement and Phase 2 Proposal.

Phase 2: Execution & Validation (Months 7-36)

Task 2.1: CF3D Finalized Design and Process Development (Months 7-18)

Activity: Refine preliminary design of Phase 1 and configure the UIUC CF3D platform for 12K carbon tow (or Sponsor from Korean SME) and B-staged resin. Print trial tubes on dummy mandrels to validate steering radii <3 cm and non-geodesic stability.

Deliverable: Prototype Composite Preforms.

Task 2.2: Finalize Chemistry and Hybrid Integration & FP Curing (Months 12-24)

Activity: Refine and down select chemistry formulations explored in Phase 1 and integrate functionalized metal liners with CF3D preforms. Inject DCPD resin and trigger Frontal Polymerization. Monitor thermal wave propagation using IR thermography to ensure uniform cure.

Deliverable: Full-scale hybrid tube prototypes (Straight or Branched joint).

Task 2.3: Cryogenic Testing & Desorption Cooling (Months 18-30)

Activity: Work with Cryogenics Test Laboratory (CTL) to install prototypes in the Cryostat-100 facility. Perform liquid nitrogen LN_2 chill-down tests to validate thermal mass models. Simulate ascent depressurization to measure the cooling power of the aerogel layer.

Deliverable: Thermal performance data report confirming heat leak <1W/m and successful desorption cooling.

Task 2.4: Mechanical Validation (Months 24-36)

Activity: Work with selected testing lab to perform burst pressure testing. Investigate microcracking and delamination via Micro-CT scan after cryogenic thermal cycling (100 cycles, 300K to 20K).

Deliverable: TRL 6 validated hybrid tube system.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**University of Illinois Urbana-Champaign (UIUC):** **Role:** Core Technology Developer & Scientific Lead. **Responsibilities:** ○ Development of the **CF3D** algorithms and printer hardware. ○ Formulation of the **DCPD resin** and **Frontal Polymerization** triggers. ○ **Computational Modeling** (FEA/CFD) of the hybrid structure. ○ **Cryogenic Testing** at the Cryogenics Test Laboratory (CTL). Design reviews, integration workshops, and publication of results in high-impact journals
Expected R&R of Korean SME Partner	**Role:** Industrial Scaling & Component Supplier. **Responsibilities:** ○ **Liner Fabrication:** Manufacturing high-precision thin-walled Aluminum/Titanium liners using industrial hydroforming or welding processes. ○ **Aerogel Supply:** Providing industrial quantities of optimized aerogel blankets (e.g., JIOS Aerogel capabilities). ○ **Commercialization:** Leading the integration of these tubes into the Korean hydrogen supply chain (mobility, fueling stations) and conducting standards compliance testing (ISO/KGS).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This Joint R&D Project executes the Convergent Manufacturing Paradigm to revolutionize Liquid Hydrogen (LH2) infrastructure, delivering a commercially viable Type IV-Hybrid Storage System. Technologically, the project targets a breakthrough in cryogenic durability, achieving a 3x increase in fracture toughness G_{IC} 1.7 N/mm to resolve persistent delamination issues. By utilizing Frontal Polymerization and advanced steering, we enable energy-efficient, rapid manufacturing of complex, leak-free geometries without heavy fittings. Additionally, integrated Desorption Cooling facilitates active thermal management and rapid fueling. Commercially, this collaboration bridges the academic-industrial gap, providing the Korean SME partner with exclusive, scalable manufacturing for the UAV, and heavy-duty truck markets. This low-CAPEX process also meets stringent aerospace requirements for next-generation launch vehicles. Broadly, the project accelerates the global hydrogen economy by lowering storage costs and fostering a resilient US-Korea supply chain. Ultimately, this initiative will deliver a TRL 6 prototype and contribute critical data to future ISO/KGS cryogenic safety standards, transforming scientific innovation into a strategic industrial asset.

6. References

1. Spond, D. E., Laintz, D. J., Hall, C. A., & Dulaigh, D. E. (1974). Vacuum jacketed composite propulsion feedlines for cryogenic launch and space vehicles. Volume 2: Appendixes (No. NASA-CR-134554).
2. Wu, Ivan C., et al. "Additive Continuous Tows Preform for Fiber Angle-Optimized Composite Tube." American Society for Composites (ASC) Annual Technical Conference. Cham: Springer Nature Switzerland, 2024.
3. Fujiwara, Takashi, et al. "Mechanical Properties of Epoxy-and Dicyclopentadiene-Based Carbon-Fiber-Reinforced Plastics at Low and Room Temperatures." Applied Composite Materials (2025): 1-13.
4. Data extracted from https://www.buyaerogel.com/product/cryogel-z/)
5. Zheyuan Zheng., et al. "Energy Efficient In Space Manufacturing of High Performance Composite Longeron", AIAA SCITECH 2026 Forum, https://doi.org/10.2514/6.2026-0615
6. Fan, Changfeng, et al. "A Simple Surface Treatment for Improving the Adhesive Bonding Properties and Durability of an AlMg3 Alloy." Molecules 29.24 (2024): 5856.

(max 10 pages)

7. Zhu, Wei, et al. "Effect of different coupling agents on interfacial properties of fibre-reinforced aluminum laminates." Materials 14.4 (2021): 1019.
8. Saif, T., Saad, N., Al-Zubiedy, A., & Idzikowski, A. (2022). Effect of ply angle on the burst pressure of composite pressure vessels by filament winding. Budownictwo o Zoptymalizowanym Potencjale Energetycznym, 11, 223-233.
9. Swanger, A. M., and MA Guerrero Nacif. "Passive cooling in aerogel-based insulation systems for liquid hydrogen upper stage launch vehicle tanks." Cryogenics 128 (2022): 103591.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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